CSSE3113 – Introduction to Software Engineering

Understanding Requirements

Requirements

IEEE Standard 729 defines it as

- “A condition or capability needed by a user to solve a problem or achieve an objective”

- “A condition or capability that must be met or possessed by a system...to satisfy a contract, standard, specification, or other formally imposed document”

Requirement Engineering

- Many software developers want to jump right in building software before they have a clear understanding of what is needed.
- They argue that things will become clear as they build, that project stakeholders will be able to understand need only after examining early iterations of the software, that things change so rapidly that any attempt to understand requirements in detail is a waste of time.
- The bottom line is producing a working program and all else is secondary.
- What makes these arguments seductive is that they contain elements of truth. But each is flawed and can lead to a failed software project.
- The broad spectrum of tasks and techniques that lead to an understanding of requirements is called *requirements engineering*.
- It beings with communication activity and continues into the modeling activity. It builds a bridge to design and construction.

Software Requirements and Requirement Engineering

- The requirements for a system
  - are the descriptions of the services provided by the system and its operational constraints.

- These requirements reflect the
  - needs of customers for a system that helps solve some problems such as controlling a device, placing an order or finding information.

- *"The process of finding out, analysing, documenting and checking these services and constraints is called requirements engineering (RE)."*

Different Levels Of Abstraction

- User requirements
  - are statements, in a natural language plus diagrams, of what services the system is expected to provide and the constraints under which it must operate. The user requirements may vary from broad statements of the system features required to detailed, precise descriptions of the system functionality.
- System requirements
  - System requirements set out the system's functions, services and operational constraints in detail.
  - The system requirements document (sometimes called a functional specification) should be defined exactly what is to be implemented.
  - It may be part of the contract between the system buyer and the software developers.

Example 1

- *User requirement:* The library system should provide a way to allow a student to borrow a book from the library.

- *System requirement:* The library system should provide a withdraw interaction that allows a student to withdraw a book given the *isbn* and copy *number* of the book to be withdrawn.

- The interaction fails if: the book is already withdrawn, the book is not in the library's collection, the student has already withdrawn 5 books, the student owes more than $5, the book is on hold by someone else. Otherwise…(*To be completed*)

Example 2

User requirements definition

1. The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System requirements specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
1.2 The system shall generate the report for printing after 17.30 on the last working day of the month.
1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
1.4 If drugs are available in different dose units (e.g. 10mg, 20mg, etc.) separate reports shall be created for each dose unit.
1.5 Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.

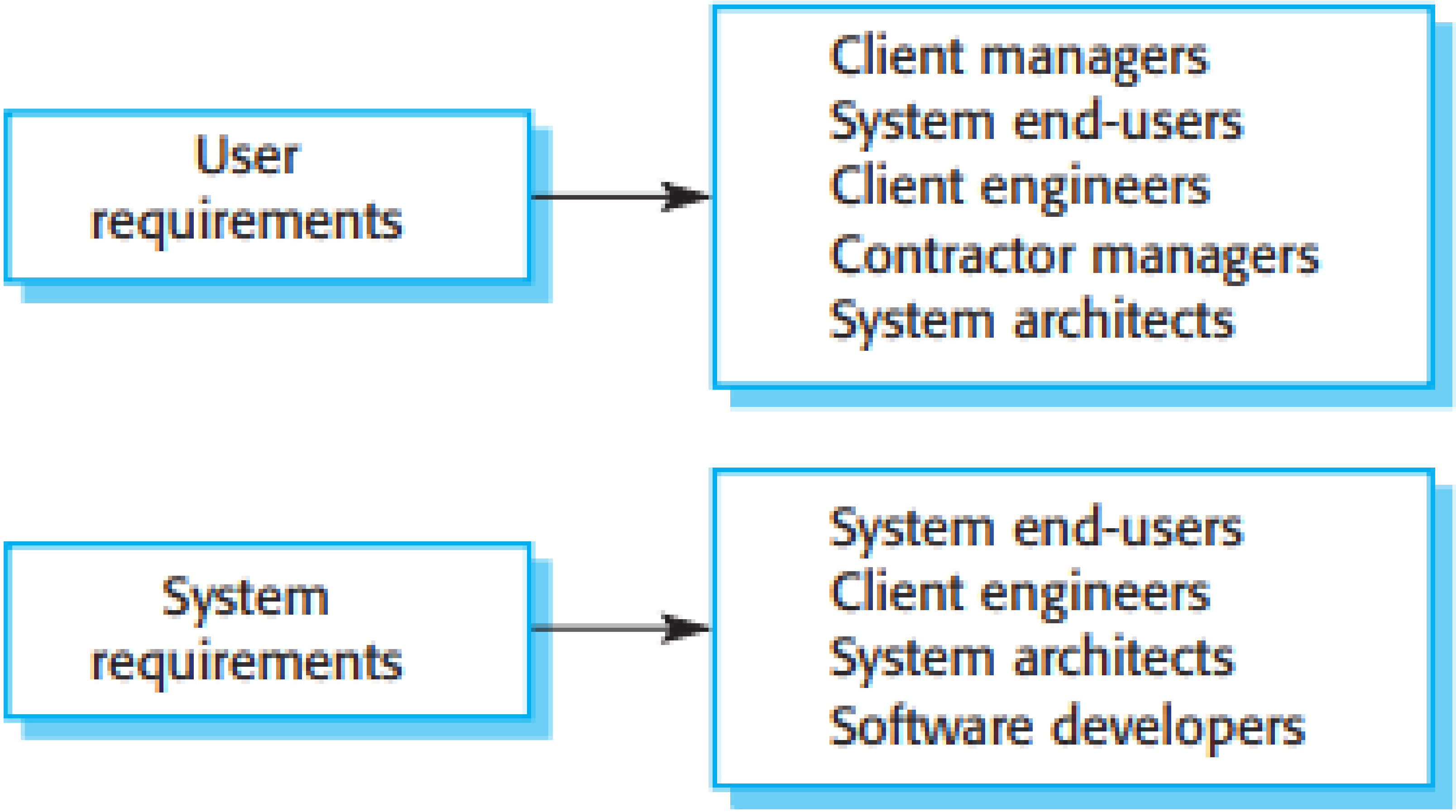
Requirements for Different Stakeholders

- You need to write requirements at different levels of detail because different types of readers use them in different ways.
- The readers of the user requirements are not usually concerned with how the system will be implemented
- Managers are not interested in the detailed facilities of the system.
- The readers of the system requirements need to know more precisely what the system will do.
- They are concerned with how it will support the business processes or because they are involved in the system implementation.

Requirements for Different Stakeholders

- The different types of document readers are actually the system stakeholders.
- System stakeholders include anyone who is affected by the system in some way and so anyone who has a legitimate interest in it.
- Stakeholders range from end-users of a system through managers to external stakeholders such as regulators, who certify the acceptability of the system.

Requirements for Different Stakeholders



Requirements for Different Stakeholders

- For example, system stakeholders for the Mentcare system include:
  - Patients whose information is recorded in the system and relatives of these patients.
  - Doctors who are responsible for assessing and treating patients.
  - Nurses who coordinate the consultations with doctors and administer some treatments.
  - Medical receptionists who manage patients' appointments.
  - IT staff who are responsible for installing and maintaining the system.
  - A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care.
  - Health care managers who obtain management information from the system.
  - Medical records staff who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.

Types Of Requirements

- Software system requirements are often classified as
  - Functional requirements
  - Non-functional requirements

Functional Requirements

- These are statements of services the system should provide.
- They bring in the system’s view and define from the system’s perspective
  - Services the system should provide
  - How the system should react to particular input
  - How the system should behave in a particular situations

  or

- What the system should not do

Functional Requirements

- Functional requirements for the Mentcare system, used to maintain information about patients receiving treatment for mental health problems:
  1. A user shall be able to search the appointments lists for all clinics.
  2. The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
  3. Each staff member using the system shall be uniquely identified by his or her eight-digit employee number.

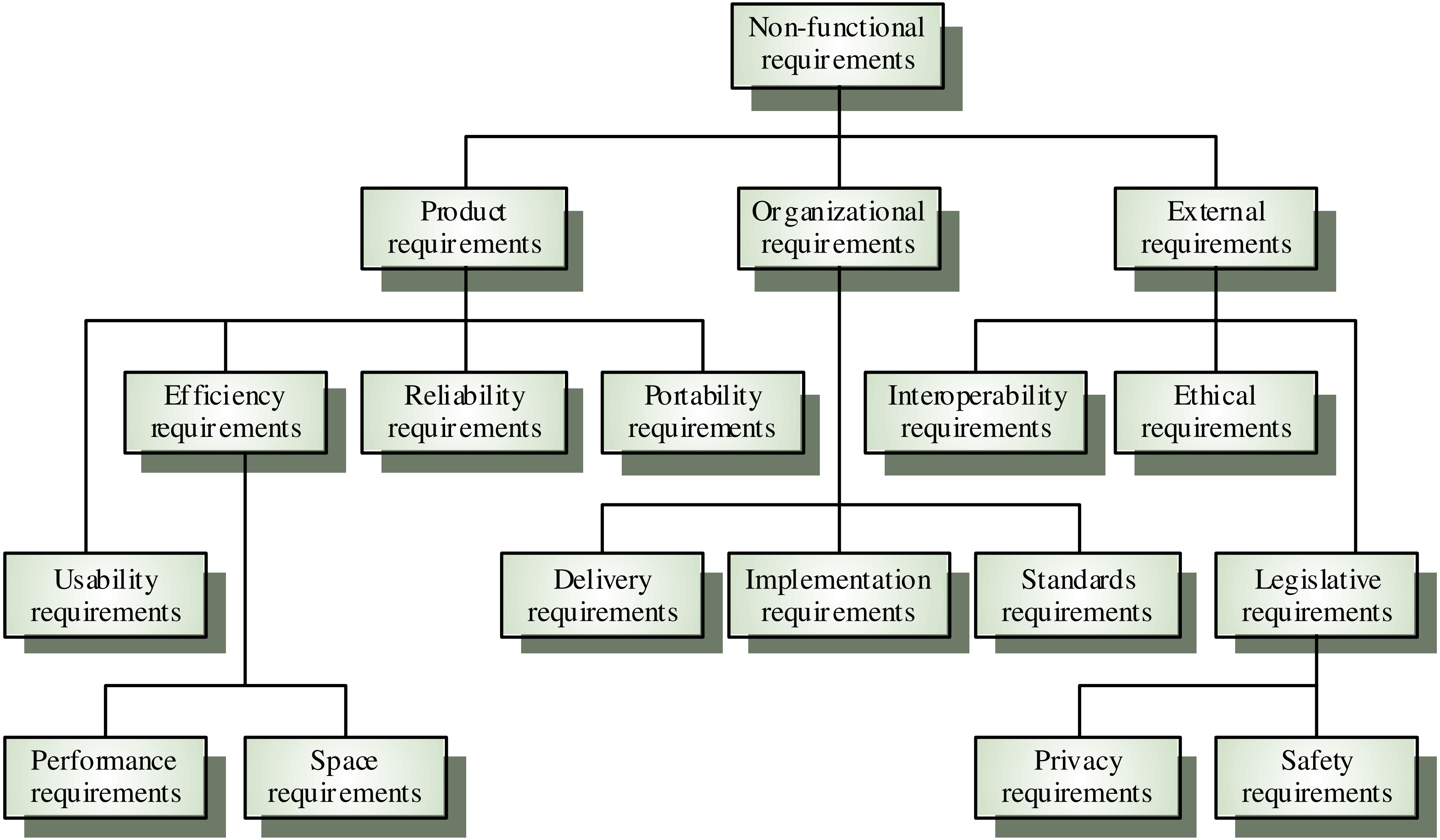
Requirements Completeness and Consistency

- **Completeness** means that all services and information required by the user should be defined.
- **Consistency** means that requirements should not be contradictory.
- In practice, it is only possible to achieve requirements consistency and completeness for very small software systems.
- One reason is that it is easy to make mistakes and omissions when writing specifications for large, complex systems.
- Second reason is that large systems have many stakeholders, with different backgrounds and expectations.
- Inconsistent requirements may only be discovered after deeper analysis or during system development.

Non-functional Requirements (NFR's)

- These are constraints on the services or functions offered by the system.
- They include timing constraints, constraints on the development process and constraints imposed by standards.
- Non-functional requirements often apply to the system as a whole. They do not usually just apply to individual system features or services.

Types of NFR's



Types of NFR's (Contd..)

- **Product requirements**
  - Requirements which specify that the delivered product must behave in a particular way.
  - e.g. execution speed, reliability, Usability, etc.
- **Organizational requirements**
  - Requirements which are a consequence of organizational policies and procedures in the customer's and developer's organization.
- **External requirements**
  - Requirements which arise from factors which are external to the system and its development process
  - e.g. legislative requirements, etc.

Types of NFR's (Contd..)

Reliability Requirements:

Reliability requirements deal with the failure to provide service.

Example :

The failure frequency of a heart-monitoring unit that will operate in a hospital's intensive care ward is required to be less than one in 20 years. Its heart attack detection function is required to have a failure rate of less than one per million cases.

Types of NFR's (Contd..)

Efficiency Requirements:

- Deals with the hardware resources needed to perform the functions of the software.
- The main hardware resources to be considered are the computer's

  processing capabilities (measured in MIPS – million instructions per second, MHz or megahertz etc.), its data storage capability in terms of memory and disk capacity (measured in MBs – megabytes, GBs – gigabytes, etc.) and the data communication capability of the communication lines (usually measured in KBPS – kilobits per second., etc)
- The requirements may include the maximum values at which the hardware resources will be applied in the developed software system

Types of NFR's (Contd..)

Portability Requirements:

Portability requirements tend to the adaptation of a software system to other environments consisting of different hardware, different operating systems, and so forth.

Interoperability Requirements:

- Interoperability describes the extent to which systems and devices can exchange data, and interpret that shared data.
- For two systems to be interoperable, they must be able to exchange data and present that data such that it can be understood by a user

Types of NFR's (Contd..)

Integrity – deal with system security that prevent unauthorized persons access.

Usability – deals with the scope of staff resources needed to train new employees and to operate the software system.

Integrity

Example

The Engineering Department of a local municipality operates a GIS (Geographic Information System). The Department is planning to allow citizens access to its GIS files through the Internet. The software requirements include the possibility of viewing and copying but not inserting changes in the maps of their assets as well as any other asset in the municipality's area ("read only" permit). Access will be denied to plans in progress and to those maps defined by the Department's head as limited access documents.

Usability

The software usability requirements document for the new help desk system initiated by a home appliance service company lists the following specifications:

(a) A staff member should be able to handle at least 60 service calls a day.

(b) Training a new employee will take no more than two days (16 training hours), immediately at the end of which the trainee will be able to handle 45 service calls a day.

Example: A Word Processor

Let us now look at an example to understand the difference between these different types of requirements.

Let us assume that we have a word-processing system that does not have a spell checker. In order to be able to sell the product, it is determined that it must have a spell checker

Contd..

- The **User Requirement** could be as follows:

  Finding spelling errors in the document and deciding whether to replace each misspelled word with one chosen from a list of suggested words. It is important to note that this requirement is written from a user's perspective.
- After documenting the user's perspective in the form of user requirements, we look at the system's perspective:
- what is the functionality provided by the system and how will it help the user to accomplish these tasks.

Contd..

The **functional requirement** for the same user requirement could be written as follows:

- *The spell checker will find and highlight misspelled words.*
- *It will then display a dialog box with suggested replacements.*
- *The user will be allowed to select from the list of suggested replacements.*
- *Upon selection it will replace the misspelled word with the selected word.*

- Finally, a **non-functional requirement** of the system could require that it must be integrated into the existing word-processor that runs on windows platform

Specifying the Requirements

- Functional and nonfunctional requirements should be specified in such a way that they are understandable by system users who don’t have detailed technical knowledge.

- These requirements are defined using natural language, tables and diagrams as these can be understood by all users.

Guidelines

- While specifying or writing the requirements, following guidelines must be followed:
  - Invent a standard format and use it for all requirements.
  - Use language in a consistent way.
  - Avoid the use of computer jargons.
  - Separate functional and non-functional requirements.
  - Distinguish requirements priorities
    - Example: MoSCoW (Must, Shall, Could, Want/Will (no TBD))
  - Testable (write test cases)

Example – Identify the problem

If sales for current month are below target sales, then report is to be printed unless difference between target sales and actual sales is less than half of difference between target sales and actual sales in previous month, or if difference between target sales and actual sales for the current month is less than 5%.

Contd..

Problems:

- Difficult to read
- Ambiguity: sales and actual sales, 5% of what?
- Incomplete: what if sales are above target sales?

Task

- Write plausible system requirements for the following functions:
- An unattended petrol (gas) pump system that includes a credit card reader. The customer swipes the card through the reader, then specifies the amount of fuel required. The fuel is delivered and the customer's account debited.